

Sensitivity analysis

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Ablation studies

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Statistical tests and assumptions

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sampling variability

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Sensitivity analysis. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Sensitivity analysis. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Thresholds .4, .5, .6 give passes8, 9, 7 out of10. Report range and a selection limit.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled sensitivity analysis evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of sensitivity analysis. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Sensitivity analysis

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

Scores are 0.4,0.6,0.8. Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

Score decisions under inclusive cutoffs

	Score 0.4	Score 0.6	Score 0.8
Cutoff 0.5	No	Yes	Yes
Cutoff 0.7	No	No	Yes
Cutoff 0.6	No	Yes	Yes

Read a cell using both its row and column labels.

Sensitivity analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Hold scores fixed

Keep the same three supplied scores.



2. Apply cutoff 0.5

0.6 and 0.8 meet the boundary.



3. Apply cutoff 0.7

Only 0.8 meets the new boundary.

CHANGE ONE THING

Set the cutoff to exactly 0.6.

Explain the changed result using the matching lesson.

Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only 0.8 meets the new boundary.

1. Hold scores fixed
2. Apply cutoff 0.7
3. Apply cutoff 0.5

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Apply cutoff 0.7 Only 0.8 meets the new boundary. Sensitivity analysis checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Practice 2: explain the deciding evidence

Changed-case practice: Set the cutoff to exactly 0.6. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. 0.6 and 0.8 are positive because the rule includes equality.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: 0.6 and 0.8 are positive because the rule includes equality. The declared at-least boundary includes the threshold value.

Repair a missing building block

Review Ablation studies (ai-7/16). Goal: Explain the mechanism and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the mechanism and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the mechanism and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
